

Daily Problem

November 24, 2025

Find:

$$\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx.$$

Solution:

$$\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx = \int \frac{\tan x}{\tan^3 x + 1} \frac{1}{\cos^2 x} dx = \int \frac{\tan x}{\tan^3 x + 1} d \tan x$$

let $t = \tan x$, we get :

$$\begin{aligned} \int \frac{\sin x}{\sin^3 x + \cos^3 x} dx &= \int \frac{t}{1+t^3} dt = \int \frac{t}{(t+1)(t^2-t+1)} dt = \int -\frac{1}{3} \frac{1}{1+t} + \frac{1}{3} \frac{t+1}{t^2-t+1} dt \\ &= -\frac{1}{3} \ln |1+t| + \int \frac{1}{3} \frac{\frac{1}{2} dt^2 - t + 1}{t^2 - t + 1} dt + \int \frac{1}{3} \frac{\frac{3}{2}}{t^2 - t + 1} dt \\ &= -\frac{1}{3} \ln |1+t| + \frac{1}{6} \ln |t^2 - t + 1| + \frac{1}{2} \int \frac{1}{(t - \frac{1}{2})^2 + \frac{3}{4}} dt \\ &= -\frac{1}{3} \ln |1+t| + \frac{1}{6} \ln |t^2 - t + 1| + \frac{1}{2} \frac{2}{\sqrt{3}} \arctan \frac{t - \frac{1}{2}}{\frac{\sqrt{3}}{2}} + C \end{aligned}$$

In conclusion:

$$\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx = -\frac{1}{3} \ln |1 + \tan x| + \frac{1}{6} \ln |\tan^2 x - \tan x + 1| + \frac{1}{\sqrt{3}} \arctan \frac{2 \tan x - 1}{\sqrt{3}} + C$$